Effect size and practical importance

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

One complete effect-definition and practical-decision record.

Keep paired observations, sign, magnitude, criterion, decision and unmeasured task requirements together.

Keep one habit: Recall → W1–W3/W5 → independent new paired-gain case → practical-effect project → delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.
- Without opening the old lesson, explain one key idea from Uncertainty intervals. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Confounders and controls. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Hypotheses and measurable outcomes. Give an example and a mistake that the idea helps you avoid.

Uncertainty intervals

> **My remembered idea, example and correction**

Confounders and controls

> **My remembered idea, example and correction**

Hypotheses and measurable outcomes

> **My remembered idea, example and correction**

W1 · Explain the current example

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Effect size describes magnitude on a declared scale; practical importance uses a task-specific criterion recorded before outcomes. Main paired baseline times 10, 12, 8 and candidate times 8, 9, 7 seconds give A–B gains 2, 3, 1, averaging 2. Positive means faster candidate. The minimum useful mean gain is at least two seconds, so equality meets it, even though Q3 gains only one second.

  Quality/resource evidence and statistical uncertainty are not supplied. Separate picture: A has twenty errors per thousand tasks and B fifteen, giving rates 2%/1.5%, five fewer errors per thousand, 0.5 percentage points and 25% relative reduction. This misses the declared ten-fewer-errors criterion. Changed B with eight errors gives twelve fewer, a 1.2-point or 60% relative reduction, meeting that criterion only.

  Absolute and relative scales differ. Neither positive change nor practical pass proves significance, causality or complete deployment readiness. Code analyzes supplied observations, not actual timings or a population-effect study.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

W2 · Trace the worked case

> Keep the supplied inputs, comparison rule and evidence together.

Complete task and evidence

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- Baseline times 10, 12, 8; candidate 8, 9, 7 seconds. Useful mean gain requires at least 2 seconds. Show the intermediate steps and give the conclusion with its scope.

Main paired case	Baseline seconds	Candidate seconds	Gain: baseline – candidate
Q1	10	8	2
Q2	12	9	3
Q3	8	7	1
Mean gain	(2+3+1)/3	2 seconds	Meets inclusive mean threshold 2
Individual variation	Q3 gain is1 second	Mean rule still met	Not a claim every case meets an individual 2-second target

Main paired case	Baseline seconds	Candidate seconds	Gain: baseline – candidate
Remaining uncertainty	Only three supplied paired cases	No interval or statistical test supplied	No population-certainty or causal claim

My reasoning and evidence

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Effect size describes magnitude on a declared scale; practical importance uses a task-specific criterion recorded before outcomes. Main paired baseline times 10, 12, 8 and candidate times 8, 9, 7 seconds give A–B gains 2, 3, 1, averaging 2. Positive means faster candidate. The minimum useful mean gain is at least two seconds, so equality meets it, even though Q3 gains only one second.

  Quality/resource evidence and statistical uncertainty are not supplied. Separate picture: A has twenty errors per thousand tasks and B fifteen, giving rates 2%/1.5%, five fewer errors per thousand, 0.5 percentage points and 25% relative reduction. This misses the declared ten-fewer-errors criterion. Changed B with eight errors gives twelve fewer, a 1.2-point or 60% relative reduction, meeting that criterion only.

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- A report omits this qualification: A meaningful sample effect may remain statistically uncertain; statistical significance can also concern a practically tiny effect.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Effect size describes magnitude on a declared scale; practical importance uses a task-specific criterion recorded before outcomes. Main paired baseline times 10, 12, 8 and candidate times 8, 9, 7 seconds give A–B gains 2, 3, 1, averaging 2. Positive means faster candidate. The minimum useful mean gain is at least two seconds, so equality meets it, even though Q3 gains only one second.

  Quality/resource evidence and statistical uncertainty are not supplied. Separate picture: A has twenty errors per thousand tasks and B fifteen, giving rates 2%/1.5%, five fewer errors per thousand, 0.5 percentage points and 25% relative reduction. This misses the declared ten-fewer-errors criterion. Changed B with eight errors gives twelve fewer, a 1.2-point or 60% relative reduction, meeting that criterion only.

  Absolute and relative scales differ. Neither positive change nor practical pass proves significance, causality or complete deployment readiness. Code analyzes supplied observations, not actual timings or a population-effect study.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Baseline times are 6, 8, 10 seconds; candidate times are 5, 7, 9 seconds on the same three cases in the same order. Define each gain as baseline minus candidate. Useful mean gain must be at least 1.5 seconds. Calculate and decide.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep the research evidence

> Preserve the question, protocol, observations, analysis and limits.

Complete task and evidence

- Add a labelled **effect size** and practical importance **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of **effect size** and practical importance.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for **Effect size** and practical importance. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for **Effect size** and practical importance. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Baseline times are 6, 8, 10 seconds; candidate times are 5, 7, 9 seconds on the same three cases in the same order. Define each gain as baseline minus candidate. Useful mean gain must be at least 1.5 seconds. Calculate and decide.

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

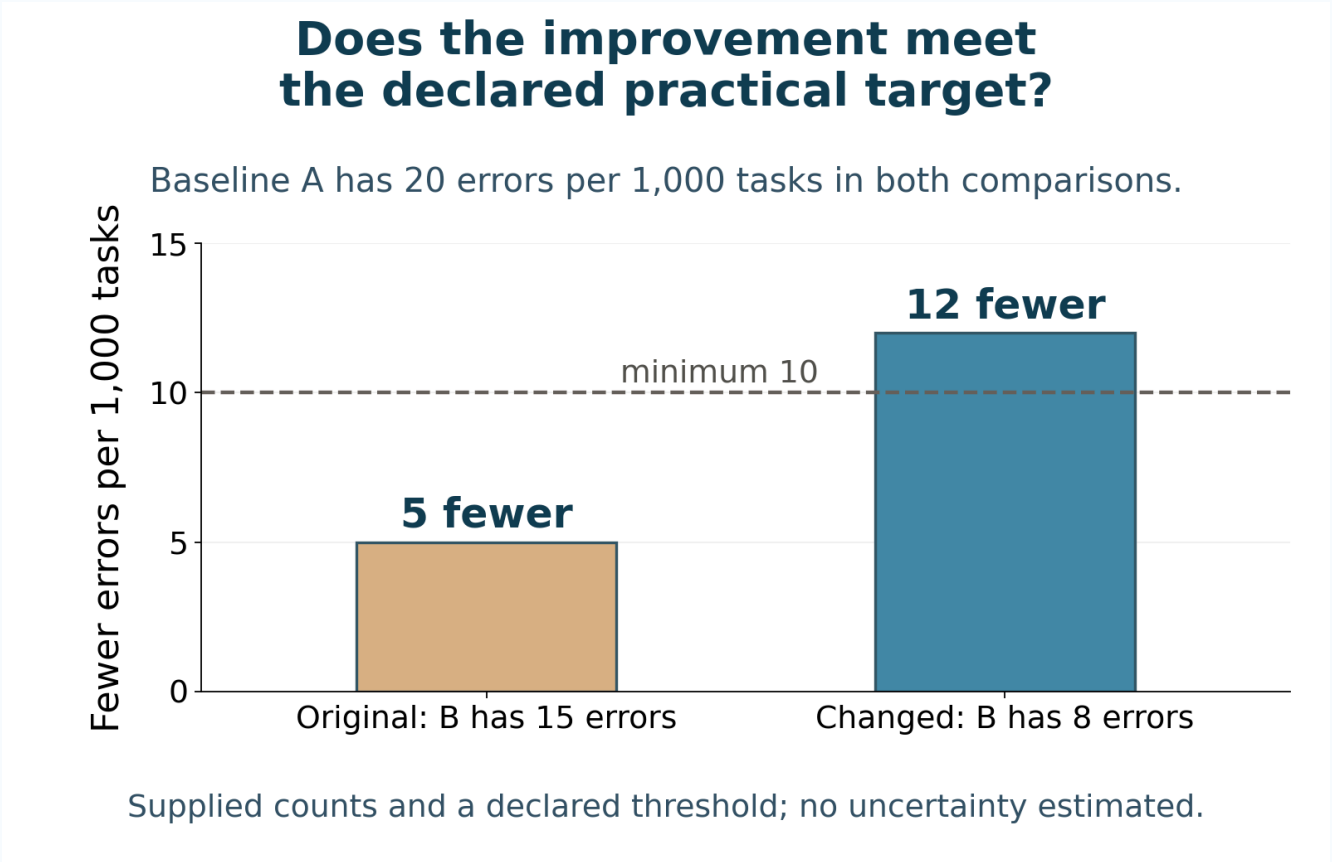
- An improvement with a **practical threshold**. A team compares error rates against a predeclared **practical threshold**.

- A makes 20 errors in 1000 tasks; B makes 15. Practical improvement requires at least 10 fewer errors per 1000.

- Calculate the absolute error-rate reduction from 20/1000 to 15/1000. Does it meet the ten-fewer-errors threshold, and what changes when B has eight errors?

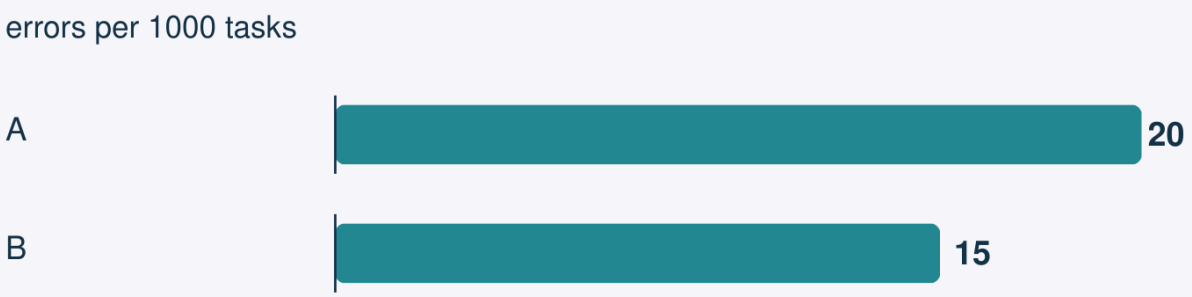
- Support used: alone / hint / worked example. This record is practice **evidence**.
- Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

  Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.



AI ACADEMY / AI-7/13

Errors on 1000 tasks



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/13

Effect size and practical importance

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compute rates

A is 2%; B is 1.5%.



2. Compute effect size

B has 5 fewer errors per 1000, a 0.5-percentage-point reduction.



3. Apply practical threshold

Five is less than ten, so the observed change misses this threshold.

CHANGE ONE THING

B makes 8 errors in 1000.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Project: Reproducible AI Research Study
- Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.
- For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep	My record
This week’s input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- An improvement with a **practical threshold**. A team compares error rates against a predeclared **practical threshold**.
- Which labelled step matches this detail? Five is less than ten, so the observed change misses this threshold.
- 1. Apply **practical threshold**
- 2. Compute rates
- 3. Compute **effect size**
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
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▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Apply **practical threshold** Five is less than ten, so the observed change misses this threshold. **Effect size** describes how much the outcome changes.

Practical importance depends on the real decision criterion; relative changes can sound large while absolute changes remain small.

Changed-case question and optional hint

- Changed-case practice: B makes 8 errors in 1000. Which conclusion is supported by the supplied facts?
- 1. The reduction becomes 12 per 1000 and meets the stated **practical threshold**.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: The reduction becomes 12 per 1000 and meets the stated **practical threshold**. Twenty minus eight exceeds the required ten fewer errors.

Repair and retain the record

Review Uncertainty intervals (ai-7/12). Goal: Explain the **mechanism** and terminology of uncertainty intervals. Short practice: Locate -0.8 and zero relative to the supplied interval [-1.4,-0.2].

Interpret the sign and confidence-method meaning, then describe what changes with interval [-1.4,0.3].. Return to this week and explain what you changed.

Review Confounders and controls (ai-7/10). Goal: Explain the **mechanism** and terminology of confounders and controls. Short practice: Draw the four condition/background combinations.

Which variables were coupled before, and why would collecting only healthy-white and spotted-blue again leave the confounding unresolved?. Return to this week and explain what you changed.

Review Hypotheses and measurable outcomes (ai-7/7). Goal: Explain the **mechanism** and terminology of hypotheses and measurable outcomes. Short practice: Name the predicted outcome and how unsupported is scored.

If B sounds friendlier but ties A's unsupported count, does the recorded improvement hypothesis receive support?. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
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▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Effect size and practical importance: the case

AI ACADEMY / AI-7/13 · 01 · LOOK AT THE FACTS

Effect size and practical importance: the case

The situation

An improvement with a practical threshold. A team compares error rates against a predeclared practical threshold.

Supplied evidence

A makes 20 errors in 1000 tasks; B makes 15. Practical improvement requires at least 10 fewer errors per 1000.

Explain effect size and practical importance using the records and boundaries of this fictional investigation.

Follow the labelled picture with the matching explanation. Open the vector diagram

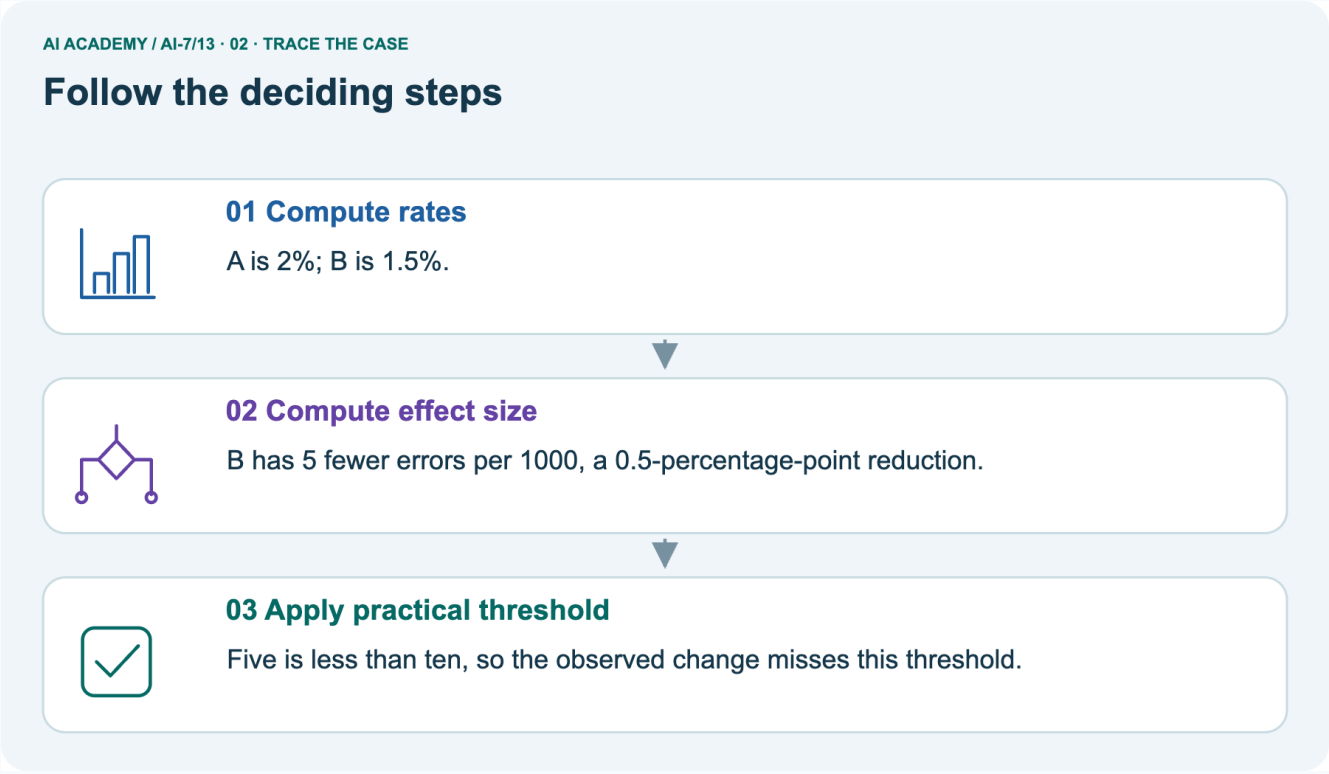
An improvement with a **practical threshold**. A team compares error rates against a predeclared **practical threshold**. Supplied facts: A makes 20 errors in 1000 tasks; B makes 15.

Practical improvement requires at least 10 fewer errors per 1000.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



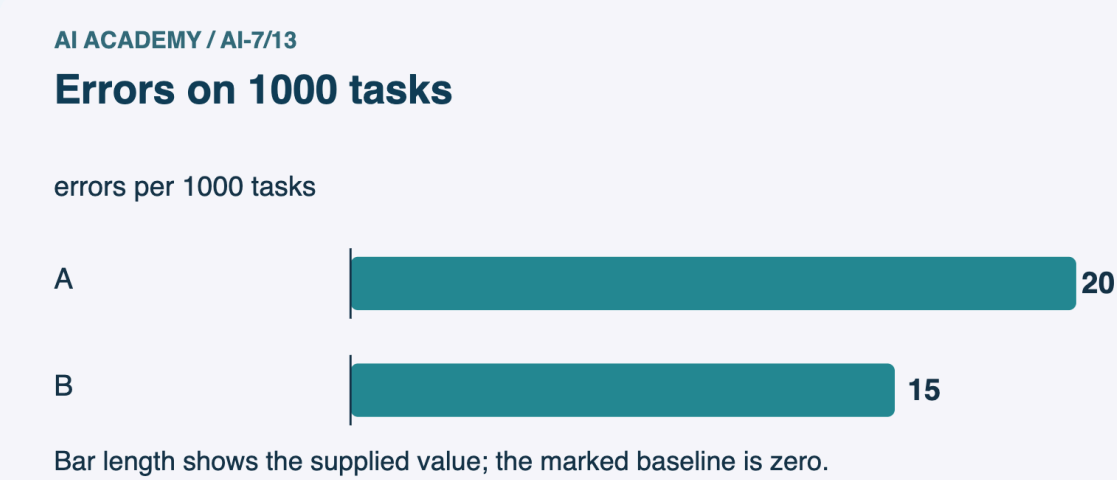
Follow the labelled picture with the matching explanation. Open the vector diagram

Compute rates: A is 2%; B is 1.5%. → Compute **effect size**: B has 5 fewer errors per 1000, a 0.5-percentage-point reduction. → Apply **practical threshold**: Five is less than ten, so the observed change misses this threshold.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Errors on 1000 tasks



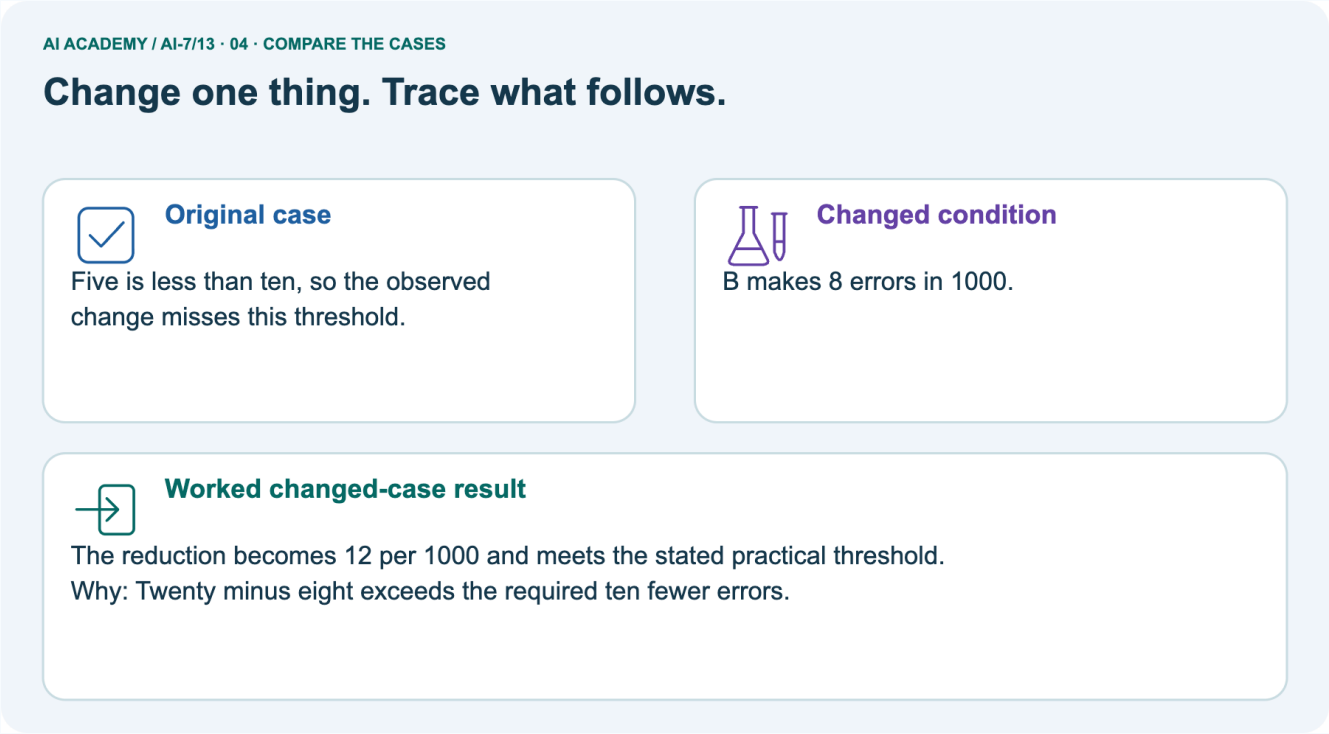
Follow the labelled picture with the matching explanation. Open the vector diagram

Errors on 1000 tasks. Units: errors per 1000 tasks. A: 20. B: 15.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

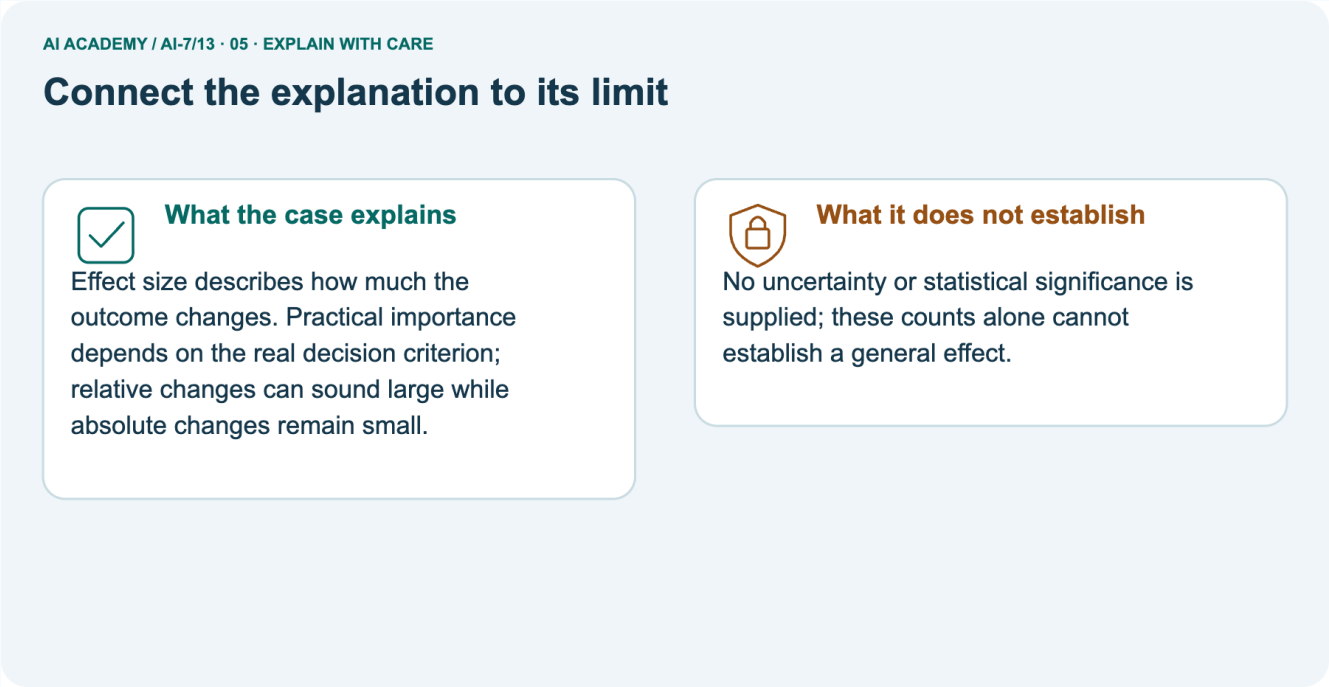
Original case: Five is less than ten, so the observed change misses this threshold. Changed condition: B makes 8 errors in 1000. Expected result: The reduction becomes 12 per 1000 and meets the stated **practical threshold**.

Reason: Twenty minus eight exceeds the required ten fewer errors.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

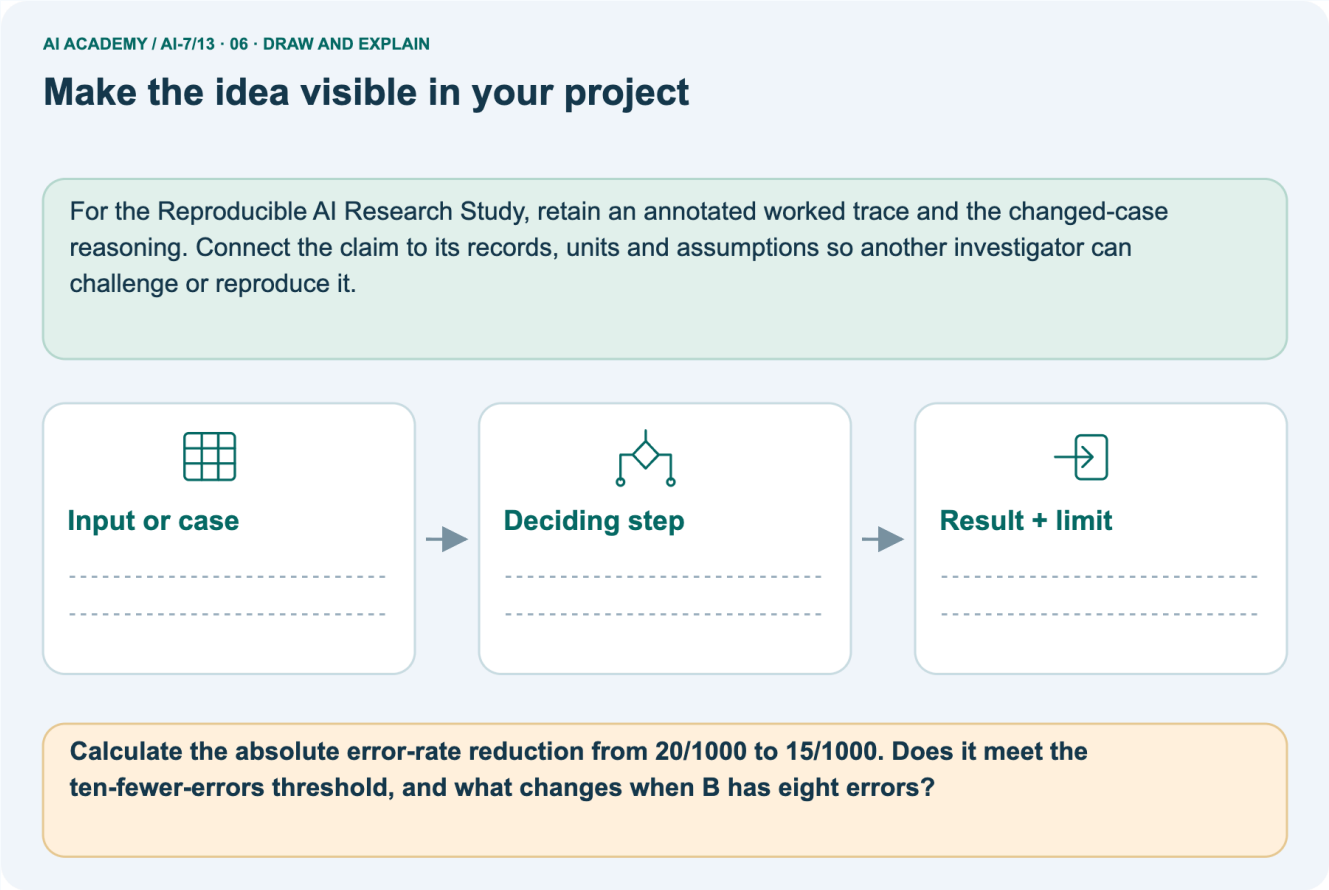
Effect size describes how much the outcome changes. Practical importance depends on the real decision criterion; relative changes can sound large while absolute changes remain small.

Limit: No uncertainty or statistical significance is supplied; these counts alone cannot establish a general effect.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate the absolute error-rate reduction from 20/1000 to 15/1000. Does it meet the ten-fewer-errors threshold, and what changes when B has eight errors?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Effect size and practical importance: the case

02. Follow the deciding steps

03. Errors on 1000 tasks

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.